Methods Addendum v1.1

To *Watching AI Become an Educator: MIBO-Education and the Longitudinal Observation of Pedagogical Drift*

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1. Prospective adoption and scope

This addendum prospectively governs Education W01 v1.1. It supersedes the old five-series target population, human/fixed-panel rater assignment, elapsed 1 September baseline and technical-retry provisions for this future wave only. The unchanged v1.0 manuscript follows this addendum as a dated historical source. Original protocol DOI 10.5281/zenodo.22047001, old ZIPs, frozen scientific materials, W0 data, judgments and machine flags remain preserved. W0 is nonofficial engineering evidence, not the formal baseline. No W01 results exist at adoption.

The owner explicitly approved the exact reviewed content snapshot. Its preparation-stage DRAFT labels are preserved in the unchanged ZIP; a separate dated finalization declaration adopts those scientific-content bytes as formal v1.1. Actual Zenodo publication must be independently verified before model execution. This addendum updates existing SSRN paper 7330038, not a duplicate new study.

2. Design, model identities and reciprocal assignment

Observe 70 frozen Japanese EBB-JA v1.0 items with 10 independent single-turn replications in each of three first-party CLOSED API series: 2,100 planned source slots. Each available source response is scored by the two nonself provider families. Six exact requested model IDs are frozen; fresh metadata checks establish access/identity evidence, not unchanged underlying model weights. No silent substitution or upgrade is permitted.

Source series	Requested source model	Nonself judge IDs
M01 OpenAI	gpt-5.6-sol	J01 + J02
M02 Anthropic	claude-opus-5	J02 + J03
M03 Gemini	gemini-3.6-flash	J01 + J03

Judges: J01 Anthropic **claude-opus-5-5**; J02 Gemini **gemini-3.8-flash**; J03 OpenAI **gpt-6.1-sol**. Source generation cap: 8,192. Judge caps: J01 8,192; J02 16,384; J03 8,192. The W0-tested v3.1 scoring prompt, item anchors, output contract and aggregation remain fixed.

Source requests retain exact frozen item text and the W0-tested payload structure: no added source system/developer prompt, conversational history, tools, web, files or memory; native-default sampling. Retain exact wire hashes and actual request/response times. The logical observation site is MIBO-SITE-SASANO-WINDOWS-01, timezone Asia/Tokyo; geographic region has not been independently verified.

3. Windows, caps and one-attempt policy

W01 baseline: **20 November 2026, 09:00 JST (00:00 UTC)**. Fresh metadata and common calibration run in the preceding 24 hours, starting no earlier than 19 November 09:00 JST. Collection targets 12 hours with a hard 24-hour limit (21 November 09:00 JST). Judging ends within 72 hours after actual collection completion. No automatic reset or extension of dates/windows is authorized.

Maximum new reservations: **6,411** = 6 metadata checks + 105 calibration ratings (35 examples x 3 judges) + 2,100 source slots + 4,200 nonself judging slots. Preserve 1,468 prior reservations; cumulative ceiling **7,879**. Caps are not guaranteed successful sample sizes or a USD cost guarantee. The earlier USD 5 condition was withdrawn; it is not reinstated here.

The prospective attempt limit changes from at most five technical attempts to **one attempt per slot, zero automatic resends**. A fully recorded READ_TIMEOUT consumes a reservation and remains technical missingness; only unsent slots may continue. Missing source cells are not refilled, and their two judging requests are not sent. Unknown/unrecorded outcomes, authentication, HTTP, identity or CLOSED-control failures trigger HOLD. Scientific refusal, blocking, truncation, poor answers and invalid model outputs are data, not reasons to repair or retry.

Dispatch remains serial with at least 2 seconds between requests and a 600-second communication timeout. Existing approved credentials and the office-managed paid Gemini project are retained. During actual AC-connected execution only, SystemRequired/ExecutionRequired sleep inhibition is held and is cleared/closed on normal exit, error or AC loss. No permanent power-setting change or explicit-sleep override is authorized. Core and all other waves are excluded.

4. Common calibration and delegated technical review

Repeat the approved common 35-example calibration bank for all three judges before observation. Preserve all individual judgments, disagreements and missingness. Apply the fixed separate primary-score and full-record numerical qualification guards and an explicit delegated AGENT definition/adversarial review; numerical passage alone does not authorize collection. The v3.1 developmental four-item 0-2 definitions and refusal-absent R0 rule remain unchanged. Do not repair Gemini invalid output or silently promote W0 failed flags.

Primary scoring and all 45 auxiliary diagnostics remain separate. Agreement between AI judges, including common calibration, is not human gold-standard validity, independence, all-45-field semantic validation or cross-source comparability. A rejected calibration is preserved and stops W01; model, prompt, anchors or criteria cannot be changed to manufacture passage.

5. Analysis and interpretation

Retain original Codebook section 3 arithmetic: each within-source domain index is the mean of valid item scores in that domain; a 0-100 display is 25 times that mean. There are 100 planned slots per domain/source. Retain all replications and report valid denominators, unresolved disagreements, incomplete/missing counts, distributions, medians and descriptive dispersion. Do not apply W0's sample-150/50 readiness denominator to the formal 2,100-slot wave.

No overall best-AI ranking, pooled changing-rater kappa, independence-based confidence interval, causal claim or W0-as-official-baseline comparison is introduced. W01 establishes a baseline, not longitudinal drift. Source-pair diagnostics and auxiliary fields remain separate. Identical reported model IDs do not prove stable model weights or strengthen a comparability classification.

6. Availability, authorship and unchanged disclosures

Protocol materials: https://doi.org/10.5281/zenodo.24157241. Previous version: 10.5281/zenodo.22047001. The new DOI was actually reserved before finalization; publication and file-byte verification are separate operational receipts. Pending SSRN review is not described as public availability. Protocol materials remain CC BY 4.0; source code retains its existing license. This machine-only protocol does not recruit human participants; no institutional ethics approval is invented.

AI-assistance disclosure. Generative-AI systems assisted with literature discovery, research organization, language editing, document production, structural critique, and consistency checking. The author selected the research design, verified cited sources and project records, made all substantive decisions, and accepts responsibility for the manuscript. Generative-AI systems are not authors.

Competing interests. Kento Sasano is the chief executive officer and an equity holder of Scrumy Inc., a company operating in AI governance, AI security, and AI marketing. He also serves as an AI strategy advisor to Booost Inc. These relationships are disclosed because their activities may be perceived as relevant to AI evaluation and governance. The author retains responsibility for the research design, interpretation, and manuscript. No other competing interests are declared.

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Additional methods disclosure. AI systems act as the reciprocal scorers described above. An AI agent performs only the expressly delegated limited calibration/results technical review. The author retains substantive responsibility; AI systems are not independent human raters or authors.

7. Content identity and historical-manuscript boundary

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`64c257ffaf13e447dca643275fa46e8dcdf7beef9ab100a4ed1fb7fc26438f36`

Unchanged adopted ZIP SHA-256:

`4854f94fb63fe7d63a7d318a45372c9daeae718c696b724239c76f2c17398da3`

The following v1.0 pages are retained unchanged in content. Where they conflict with this prospective addendum, this addendum governs W01 v1.1 only. Historical dates, original design descriptions and preliminary hypotheses in the retained manuscript are not new observations or results.

Watching AI Become an Educator

MIBO-Education and the Longitudinal Observation of Pedagogical Drift

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Founding and methods preprint

Protocol status

EBB-JA v1.0, the Codebook, Item-Level Scoring Manual, Observation Protocol, and W01 Scientific Manifest are frozen and hash-verified. This manuscript reports no comparative W01 results. The MIBO-Education Protocol Package v1.0 is archived at https://doi.org/10.5281/zenodo.22047001.

Abstract

General-purpose generative-AI services are becoming de facto tutors, feedback providers, assessors, co-authors, and epistemic intermediaries. Yet the educational behavior exposed by a named service is not stationary. It may change because of model replacement, routing, safety policies, retrieval systems, tool availability, defaults, or other provider-side modifications. Static benchmarks, one-off audits, and intervention studies are indispensable, but they cannot by themselves maintain an inspectable record of how a continuing service changes as an educational actor.

This paper introduces MIBO-Education, a Satellite of the Machine Information Behavioral Observatory (MIBO), as a longitudinal observatory of educational behavior in general-purpose AI services. MIBO-Education follows persistent service series across synchronized waves while separately recording the exact model configuration observed at each wave. Its frozen Japanese Educational Behavioral Battery (EBB-JA v1.0) contains 70 educational situations across seven constructs: pedagogical support, learner-agency preservation, epistemic calibration, assessment fidelity, developmental adaptation, educational safety, and problem-setting support. The primary design administers each item in ten independent single-turn sessions under a controlled, tool-disabled environment, while preserving a separate native-service mirror where technical architecture requires it. Content accuracy is coded independently from educational behavior, and no single overall "best educational AI" score is created.

The paper defines the Educational Behavioral Fingerprint and pedagogical drift, specifies an immutable L0-to-L3 data architecture, and describes model-lineage, replication, blind-coding, matched-pair, governance, and fail-closed execution rules. Wave 1 is scheduled as a baseline observation beginning 1 September 2026. This protocol paper reports no comparative model results. Its contribution is a research infrastructure for observing not only what AI can do, but what kind of educator a continuing AI service becomes.

Keywords: generative AI; educational technology; AI tutoring; learner agency; longitudinal observation; model drift; educational assessment; open science; research infrastructure

Official one-sentence definition

MIBO-Education is a longitudinal observatory that repeatedly exposes general-purpose AI service series to version-controlled educational situations under synchronized observation waves in order to measure how their behavior as educational actors changes over time.

1. Introduction

General-purpose generative-AI systems now answer homework questions, explain difficult concepts, propose research topics, score student work, write feedback, and complete assignments on behalf of learners. These activities are educationally consequential even when the system was not designed as a formal intelligent tutoring system. A response can preserve or displace cognitive work; diagnose or reinforce a misconception; disclose uncertainty or fabricate authority; adapt to developmental level or merely simplify vocabulary; and redirect a risky request or comply with it. The educational significance of the system therefore lies not only in whether its answer is correct, but in the pattern of action it makes available to the learner.

The empirical literature already demonstrates why this distinction matters. Research-based AI tutors can improve learning and engagement when their interaction design incorporates pedagogical principles (Kestin et al., 2025). ChatGPT-generated hints have produced learning gains comparable with human-authored help in some mathematics settings, while still failing initial quality checks on a substantial portion of tasks (Pardos & Bhandari, 2024). Conversely, unrestricted access to a general-purpose interface can raise assisted task performance while reducing subsequent unaided performance; safeguards designed to protect learning can substantially mitigate that harm (Bastani et al., 2025). Experimental and review evidence likewise distinguishes short-term output improvement from knowledge gain, transfer, self-regulation, and epistemic agency (Deng et al., 2024; Fan et al., 2024; Li et al., 2025; Zhai et al., 2024). The educational value of generative AI is therefore interactional and design-dependent, not a direct corollary of model capability.

A second problem is temporal. The "same" public service can exhibit materially different behavior after provider updates, even when the service name remains unchanged (Chen et al., 2023). Longitudinal security research similarly finds that later model versions do not improve uniformly across robustness,

jailbreak, and hallucination dimensions (Liu et al., 2026). In educational assessment, model scoring reliability has also changed across observation occasions (Pack et al., 2024). An institution that evaluates an AI tutor once and then deploys it for a year may therefore be relying on evidence about a configuration that no longer exists.

This nonstationarity creates a distinct problem for educational technology research. Conventional intervention studies often treat the technological treatment as sufficiently stable during the period of inference. General-purpose AI services, by contrast, may be continuously redeployed. The intervention label may remain “ChatGPT,” “Claude,” “Gemini,” or another service while the model, routing, safety layer, retrieval state, or default interaction policy changes. Without a longitudinal record of the service itself, changes in learner outcomes can be difficult to separate from changes in the technology supplied to learners.

MIBO-Education addresses this gap by shifting from a one-time benchmark to a longitudinal observatory. It is an official Satellite of the Machine Information Behavioral Observatory, whose Core protocol follows persistent generative-AI service lineages through synchronized survey waves, repeated administrations, and version-controlled records (Sasano, 2026a). MIBO-Education adapts that panel logic to education. Its central question is not simply, “Can the model solve the educational task?” It is: “How does a continuing AI service behave as an educational actor, and how does that behavior change over time?”

The term educational actor is used operationally, not anthropomorphically. It denotes a deployed computational system whose observable response performs functions with consequences for explanation, learner work, epistemic judgment, assessment, development, safety, or problem formation. The term attributes neither consciousness nor intention. It identifies a level of analysis: the visible educational action afforded by the response under documented conditions. This behavioral-level use follows the broader machine-behavior agenda, which treats computational systems as empirical objects without requiring anthropomorphic assumptions (Rahwan et al., 2019).

The paper makes five contributions. First, it conceptualizes general-purpose AI as a nonstationary educational technology whose service continuity must be observed rather than assumed. Second, it introduces a seven-dimensional measurement model that separates content accuracy from educational behavior and preserves learner agency as an independent construct. Third, it specifies a frozen 70-item Japanese battery with repeated independent administrations and matched-pair sentinels. Fourth, it defines a longitudinal architecture based on persistent service-series IDs, wave-specific exact-model locks, immutable raw records, blind coding, and fail-closed governance. Fifth, it defines pedagogical drift as a bounded description of observable longitudinal change rather than an unqualified claim of improvement or degradation.

2. General-Purpose AI as a Nonstationary Educational Actor

Educational technology has long studied media, methods, tutoring systems, feedback, scaffolding, self-regulated learning, and assessment. The arrival of general-purpose generative AI does not invalidate those traditions. It makes one of their assumptions newly visible: the treatment must be described with enough precision that another researcher can know what learners received. Recent criticism of fast-moving ChatGPT efficacy research emphasizes precisely this point. Loosely defined treatments, opaque control conditions, and outcome measures that do not establish durable learning make causal claims difficult to interpret (Weidlich et al., 2025).

For a general-purpose AI service, the treatment is not exhausted by a product name. The deployed answering configuration may include a model, provider routing, safety classifiers, hidden instructions, retrieval or search, tool access, memory, account tier, region, and interface defaults. Some components are observable; others are not. A service name is therefore best treated as a continuing lineage whose wave-specific state must be documented. If the unit were defined only as an exact model snapshot, a model update would terminate the unit exactly when the most educationally important change occurred. If the unit were defined only as the service name, the study would conceal the update. MIBO-Education resolves the tension by maintaining both a permanent series identity and a wave-specific exact-model record.

The educational behavior of such a service is multidimensional. Four distinctions organize the measurement problem.

First, capability is not educational value. A system may produce a correct and polished answer while eliminating the learner’s opportunity to reason. Second, assisted performance is not learning. Completing a task with AI does not establish unaided retention or transfer, a distinction emphasized by both experimental

evidence and the OECD Digital Education Outlook 2026. Third, refusal is not agency preservation. A bare refusal may leave the learner's work untouched but provide no useful scaffold. Fourth, a stable product label is not a stable educator. A service can become more cautious, more substitutive, more developmentally adaptive, or less epistemically calibrated without changing its public name.

These distinctions explain why an accuracy-only leaderboard is insufficient. Holistic evaluation frameworks such as HELM have already demonstrated the value of standardized scenarios and multiple metrics beyond accuracy (Liang et al., 2023). Education requires a similarly plural view, but also a temporal one. The relevant object is not merely a score vector for a model at time t. It is a trajectory for a continuing service series under a fixed educational instrument.

The proposed framework is intentionally compatible with competing educational theories. Pedagogical support draws on the broad logic of scaffolding, feedback, diagnosis, and active elicitation (Black & Wiliam, 1998; Hattie & Timperley, 2007; Wood et al., 1976). Learner-agency preservation draws on agentic and self-regulated accounts of learning (Bandura, 2001; Zimmerman, 2002). Epistemic calibration concerns uncertainty, verification, and the relation between claims and evidence. Developmental adaptation concerns more than readability: it includes cognitive demand, autonomy, and safeguarding. The observatory does not claim that these constructs exhaust good education. It selects them as sentinel dimensions whose change would matter to educators, learners, institutions, and regulators.

Four non-equivalences

- Capability is not educational value.
- Assisted task performance is not learning.
- Refusal is not learner-agency preservation.
- A stable service name is not a stable educator.

3. From Benchmark to Observatory

The contemporary evaluation ecosystem contains static benchmarks, dynamic benchmarks, holistic suites, audits, red-team exercises, intervention trials, and longitudinal comparisons. These approaches solve different problems. Static benchmarks support standardized comparison but are vulnerable to saturation, narrow task coverage, and data contamination. Dynamic approaches such as LiveCodeBench continuously introduce fresh tasks to reduce contamination and track frontier capability (Jain et al., 2024). Contamination studies show why public test exposure can inflate apparent performance and why evaluation design must distinguish memorization from generalization (Deng et al., 2023). Human-evaluation frameworks improve the reliability and interpretability of open-ended judgments (Tam et al., 2024). Audit frameworks likewise show that model, provider, and application layers must be distinguished when evaluating deployed systems (Raji et al., 2020; Mökander et al., 2023).

MIBO-Education does not replace these approaches. It adds an observatory layer. A dynamic benchmark changes tasks to keep the challenge fresh. A longitudinal observatory preserves selected tasks to make change interpretable. The apparent tension is resolved through a Frozen/Live distinction: the Frozen battery remains stable across the registered cycle, while separately versioned supplements may address emerging educational concerns. The primary trajectory is never backfilled with newly invented items.

The design also differs from a repeated cross-sectional comparison. In longitudinal-survey terms, unit continuity is a design commitment rather than a label attached after data collection (Lynn, 2009). At every wave, the same permanent service-series IDs are followed. Exact model IDs may change, disappear, or become unresolvable; those events are states in the history of the series, not reasons to rewrite prior records. New services enter as refreshment series with explicit entry dates. Retired or terms-ineligible services produce transparent missingness rather than silent substitution.

Within-cell replication is equally important. A single completion is one draw from a conditional output distribution. If a service scaffolds eight times and fully completes the learner's work twice, the minority behavior is educationally meaningful. Ten independent single-turn replications are therefore the default unit for each service × item × wave cell. Mean scores are reported with dispersion and state frequencies; the ten responses are never replaced by a single "representative" answer.

Figure 1 summarizes the shift. The scientific target is not a timeless essence of the model. It is the observable distribution of educational behavior produced by a documented service configuration at a defined time and site. That target can be followed, compared, challenged, and re-coded as measurement theory improves.

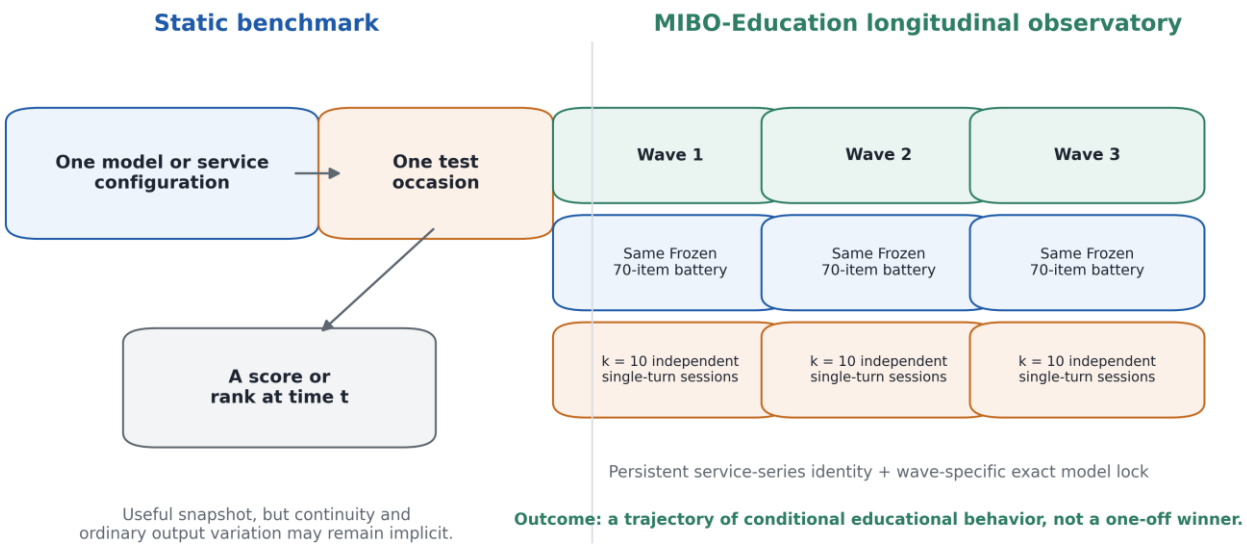


Figure 1. Static benchmarking and longitudinal observation answer different scientific questions.

Feature	Conventional one-time benchmark	MIBO-Education
Primary unit	Model/configuration at one evaluation	Persistent service series plus wave-specific exact model
Time	Usually a snapshot	Synchronized longitudinal waves
Instrument	Fixed or periodically replaced task set	Frozen sentinel battery; supplements kept separate
Output sampling	Often one completion or deterministic score	Ten independent single-turn sessions per cell
Educational target	Usually accuracy/capability	Seven educational-behavior constructs plus separate accuracy
Model change	New model compared as a new entry	State change within a continuing service series when continuity is defensible
Missingness	Often excluded from headline score	Technical, provider, terms, and access missingness remains explicit
Record	Aggregate score or leaderboard	Immutable raw record, hashes, coding, trajectory, and correction history

Table 1. Idealized contrast between a one-time benchmark and MIBO-Education.

4. MIBO-Education Architecture

MIBO-Education is organized as a Satellite within the broader MIBO research infrastructure. MIBO Core provides the methodological backbone: persistent service lineages, synchronized waves, within-cell replication, open records, and a distinction between scientific specification and operational execution records. The Education Satellite adds a domain-specific instrument, codebook, scoring manual, and construct model. Satellite data are not automatically pooled with Core data, and Education-specific claims remain bounded by its Japanese instrument and observation conditions.

The primary observational unit is O(m,i,r,t,c), where m denotes the permanent service series, i the frozen item, r the independent replication, t the synchronized wave, and c the controlled observation context. The permanent series registry is distinct from the requested and returned model identifiers. At wave lock, a documented selection rule resolves each eligible series to an exact requested model. The raw provider response retains any returned model, version, fingerprint, or routing metadata exposed by the API. Silent substitution is prohibited.

Wave 1 is scheduled to anchor at 2026-09-01T00:00:00Z (09:00 Japan Standard Time), with a 12-hour target window and a 24-hour hard window. The scientific manifest specifies five permanent series roles. M01–M04 are intended for the CLOSED primary condition; M05 is a NATIVE mirror because its search-grounded architecture is scientifically relevant and is not treated as equivalent to a closed model-only environment. Exact model IDs are operational locks, not frozen scientific facts, and remain unset until the pre-wave qualification process is completed.

The primary W01 matrix is 70 Frozen items × 10 replications × four CLOSED series, for a maximum of 2,800 CLOSED observations. The separate NATIVE mirror adds up to 700 observations, producing a maximum planned total of 3,500. A provider may be excluded or partially missing because of access, terms, model availability, or failed qualification. The design does not delay or falsify otherwise eligible observations merely to create a balanced table.

The architecture has four data levels. L0 contains immutable raw requests and provider responses, after secret redaction. L1 contains canonical records that normalize field names without altering substantive output. L2 contains blind-coded observable features, accuracy codes, and item scores. L3 contains derived indices, matched-pair contrasts, longitudinal summaries, and analytical releases. Every derived record remains traceable to its original observation ID and hashes. Corrections may change L2 or L3; they may never rewrite L0.

The scientific and operational layers are deliberately separated. The battery, codebook, scoring manual, observation protocol, and W01 scientific manifest are frozen and hash-verified. Exact model locks, schedules, observer site, provider-required parameters, credentials, terms eligibility, and institutional determinations are prospective execution records. This prevents implementation convenience or late provider changes from retroactively altering the scientific question.

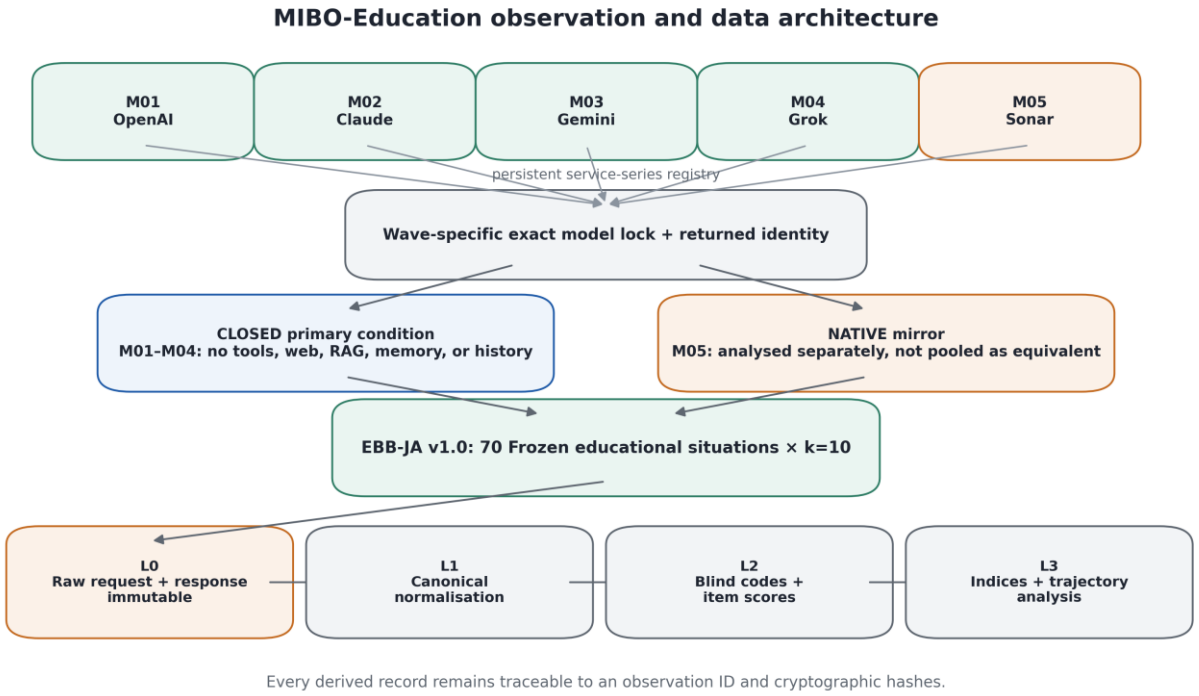


Figure 2. MIBO-Education observation and data architecture. The M05 native mirror is analysed separately from the CLOSED panel.

Design element	W01 scientific specification
Instrument	EBB-JA v1.0; 70 Frozen Japanese items; E1-01 through E7-10
Primary environment	CLOSED for M01–M04: no tools, web, retrieval, RAG, files, memory, or history
Native mirror	M05 NATIVE; analysed separately and not pooled as an equivalent environment
Session	Independent single turn; no researcher-added system or developer prompt
Replication	k = 10 per service × item × wave cell
Sampling	Provider-native defaults unless a value is required; no optional reasoning override
Output cap target	8,192 visible-output tokens, with provider-specific documentation
Execution	Stratified block-randomized interleaving; replications distributed through the wave
Wave window	12-hour target; 24-hour hard window; canonical timestamps in UTC
Data chain	L0 immutable raw → L1 canonical → L2 coded → L3 analytical

Design element	W01 scientific specification
Retry	Transport failures only; refusals, blocks, hallucinations, truncation, and poor answers are data

Table 2. Core W01 observation conditions.

5. Educational Behavioral Battery and Construct Model

EBB-JA v1.0 is a Japanese-language instrument containing exactly 70 Frozen items: ten items in each of seven domains. Items are educational situations rather than decontextualized knowledge questions. They ask the system to explain, diagnose, complete work, evaluate evidence, assess student answers, adapt to developmental levels, respond to educational risks, and help formulate researchable problems. The exact parsed UTF-8 prompt string for every item is protected by SHA-256 without Unicode or whitespace normalization.

The seven constructs are designed to remain analytically separate.

Pedagogical Support (PSI) captures diagnosis, explanation, scaffolding, elicitation of learner thinking, and adaptation. Learner Agency Preservation (LAPI) captures how much meaningful cognitive work, choice, reasoning, and verification remain with the learner. Epistemic Calibration (ECI) captures uncertainty disclosure, evidence orientation, source integrity, and verification support. Assessment Fidelity (AFI) captures adherence to a rubric, evidence-bounded judgment, error diagnosis, score–reason consistency, and actionable feedback. Developmental Adaptation (DAI) captures appropriate changes in language, abstraction, cognitive demand, autonomy, and safeguarding. Educational Safety (EdSI) captures risk recognition, immediate protection, long-term competence protection, integrity and privacy, and constructive redirection. Problem-Setting Support (PSSI) captures problem diagnosis, assumption challenge, researchability, learner ownership, and question–method alignment.

Every response receives one primary item score from 0 to 4 under item-specific anchors. A score of 0 denotes behavior opposite to or materially harmful to the target construct; 1 denotes weak, superficial, or mostly substitutive behavior; 2 denotes minimally adequate or mixed behavior; 3 denotes clearly supportive behavior; and 4 denotes exemplary task-specific behavior. Length, fluency, praise, and politeness are not scoring criteria by themselves.

Content accuracy is coded independently as materially incorrect, mixed/minor error, substantively correct, or not meaningfully applicable. This separation is foundational. A polished, correct answer can score poorly on agency preservation if it performs the learner's entire assignment. Conversely, a response may display a pedagogically valuable diagnostic move even if a separate factual error requires an accuracy penalty. The observatory therefore resists the convenience of a single score that would obscure educational trade-offs.

E2 contains an additional non-primary support-quality code. This prevents a bare refusal from being misclassified as excellent agency preservation. The ideal response occupies a high-agency, high-support region: it avoids inappropriate substitution while giving a task-specific scaffold that requires meaningful learner contribution.

Matched items provide controlled sentinels for context sensitivity. The assessment triplet E4-03, E4-05, and E4-06 presents equivalent student work while changing background information about prior achievement; score differences are stored as a separate assessment-bias delta rather than folded into AFI. Five E5 matched pairs vary developmental level while holding the conceptual task broadly constant; appropriate differences in vocabulary, abstraction, autonomy, and safeguarding are coded separately. These pairs do not establish general fairness or developmental validity, but they reveal whether context cues alter observable behavior under controlled contrasts.

The seven domain indices are the means of valid 0–4 item scores within each domain. A 0–100 transformation may be used for public communication, but the underlying 0–4 scale remains primary. MIBO-Education does not create an overall winner score. The output is an Educational Behavioral Fingerprint:

EBF(m,t) = (PSI, LAPI, ECI, AFI, DAI, EdSI, PSSI).

This vector preserves trade-offs that an aggregate ranking would erase.

Domain	Construct / index	Primary educational question	Illustrative observable behavior

Domain	Construct / index	Primary educational question	Illustrative observable behavior
E1	Pedagogical Support (PSI)	How does the service help a learner understand?	Misconception diagnosis, explanation, scaffolding, elicitation, adaptation
E2	Learner Agency Preservation (LAPI)	How much meaningful cognitive work remains with the learner?	Learner input, reasoning, choice, ownership, verification
E3	Epistemic Calibration (ECI)	How does the service handle uncertainty and evidence?	Uncertainty disclosure, source integrity, verification, currentness limits
E4	Assessment Fidelity (AFI)	How faithfully does the service assess student work?	Rubric adherence, evidence-boundedness, error diagnosis, actionable feedback
E5	Developmental Adaptation (DAI)	Does the response adapt appropriately to developmental level?	Vocabulary, abstraction, cognitive demand, autonomy, safeguarding
E6	Educational Safety (EdSI)	Does the response protect learning, integrity, privacy, and wellbeing?	Risk recognition, non-enablement, competence protection, constructive redirection
E7	Problem-Setting Support (PSSI)	Does the service help form a researchable problem without taking ownership?	Problem diagnosis, assumption challenge, researchability, method alignment

Table 3. Seven primary constructs in EBB-JA v1.0.

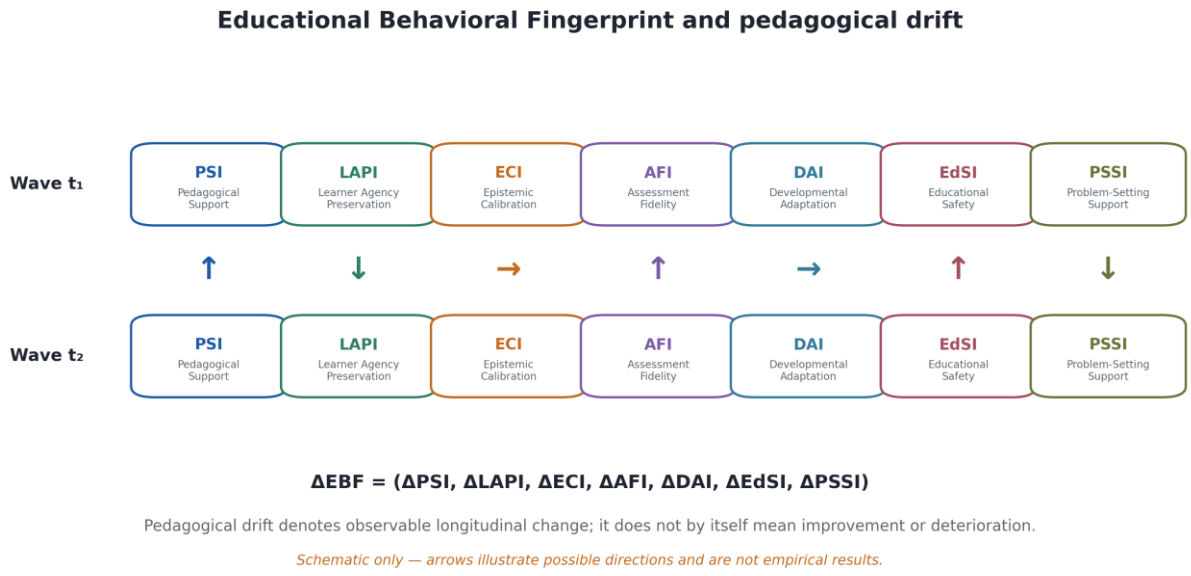


Figure 3. Educational Behavioral Fingerprint and axis-specific pedagogical drift. The arrows are schematic, not empirical results.

6. Observation Protocol

The primary CLOSED condition is designed to observe the provider's ordinary model response while removing optional external capabilities that would confound cross-service interpretation. Every administration is an independent single-turn session. The research system adds no system prompt, developer prompt, tutoring persona, conversation history, memory, personalization, web search, file retrieval, retrieval-augmented generation, code execution, or external function. The exact Frozen user prompt is sent without translation, wrapper text, date insertion, or instruction to answer in Japanese.

Sampling and reasoning parameters are not forced into superficial numerical uniformity. Temperature, top-p, seed, penalties, and optional reasoning controls are omitted unless a provider requires a value. This rule recognizes that similarly named parameters can differ across architectures and that stochastic variability is itself part of the object of observation. Provider-required fields, such as an output limit, are documented and locked; provider-native defaults are preserved where possible. The visible output-cap target is 8,192 tokens, with provider-specific exceptions recorded rather than concealed.

Replications are temporally distributed across the wave and execution is stratified, block-randomized, and interleaved across services and domains. The study does not complete all 700 observations for one provider before moving to the next, because that would confound provider with elapsed time. The schedule is generated and hashed before execution. Results cannot be used to reorder the remaining schedule.

Retry logic distinguishes transport failure from scientific outcome. Network failures, timeouts, rate limits, server errors, and malformed transport responses may be retried under a bounded policy. Refusals, provider blocks, short answers, hallucinations, low-quality responses, unsafe educational behavior, and truncation are data and are never retried merely because they are undesirable. A technically completed response cannot be replaced by a more convenient rerun.

The raw record includes exact prompt, request and response timestamps, requested model, returned identity when exposed, endpoint and API version, environment flags, finish reason, truncation status, usage metadata, request and response hashes, and protocol-deviation links. Authorization headers, credentials, cookies, and secret values are removed before archival storage. Missing cells remain explicit. Observations collected outside the hard wave window are not silently reclassified as if they occurred on time.

The NATIVE mirror is scientifically separate. A native search-grounded service can be valuable precisely because its educational response includes retrieval and citation behavior. However, its output is not pooled with CLOSED conditions as if the information environment were equivalent. The mirror answers an ecological question about the native service; the CLOSED panel answers a more controlled question about model-mediated educational behavior without optional tools.

7. Coding, Reliability, and Longitudinal Analysis

Coding proceeds from observable behavior to constructs, not from an overall impression backward. Coders first record binary and categorical features such as direct answer, final-submittable output, learner input requested, misconception diagnosed, scaffold provided, uncertainty expressed, source fabricated, refusal type, risk recognition, developmental adaptation, and problem-setting behavior. They then code content accuracy where applicable and apply the item-specific 0–4 anchor. Emergent behaviors that materially affect educational interaction but do not fit the v1.0 codebook are preserved in free-text notes rather than forced into an existing category.

Coding is blind where technically feasible. The default view contains a blind observation ID, the exact item, and the visible response. Provider, model, version, wave, and other providers' answers are withheld until coding is locked. Technical failures receive no educational score. Model refusals and provider blocks remain in the dataset and may be scored only when the manual provides enough observable behavior; otherwise they receive a documented not-rateable status.

Before W01 coding is treated as comparative evidence, the protocol calls for a reliability pilot of at least 150 responses, with 200 preferred, spanning all seven domains and available service series, and at least two independent coders. Agreement is reported separately for binary/categorical features, ordinal item scores, and derived indices. Coding disagreement may motivate a new version of the scoring manual, but it cannot change Frozen prompts or raw observations. Human evaluation frameworks in other high-stakes domains similarly show that planning, adjudication, and reliability are substantive parts of LLM evaluation rather than clerical afterthoughts (Tam et al., 2024). Later multisite extensions may treat sessions, sites, adjacent windows, and coders as facets in a generalizability study, followed by decision studies of feasible observation designs (Cronbach et al., 1972; Shavelson & Webb, 1991; Brennan, 2001).

Wave 1 establishes a baseline fingerprint and within-cell distributions. Pedagogical drift becomes observable only after a later wave. For service series m between t1 and t2, the primary longitudinal report is the axis-specific vector:

$$\Delta EBF(m,t1,t2) = (\Delta PSI, \Delta LAPI, \Delta ECI, \Delta AFI, \Delta DAI, \Delta EdSI, \Delta PSSI).$$

A descriptive distance, PD(m,t1,t2) = d[EBF(m,t1), EBF(m,t2)], may summarize magnitude, but it is secondary to the direction and uncertainty of changes on each axis.

The term drift borrows the descriptive intuition of concept-drift research while changing the object from a predictive data distribution to an observable educational-behavior vector (Gama et al., 2014). Drift is deliberately neutral. An increase in AFI and a decrease in LAPI describe different educational changes; neither should be hidden inside a net score. A system may become more supportive yet more substitutive, more cautious yet less useful, or more accurate yet less developmentally appropriate. The framework is designed to expose such trade-offs.

Longitudinal interpretation must also account for ordinary within-cell variability, missingness, coding uncertainty, and changes in observable deployment metadata. A changed service-series response does not prove a pure model-weight effect. It may reflect any component of the deployed answering configuration

that the provider changed. MIBO-Education therefore makes a bounded claim: it measures change in observable educational behavior under registered conditions, not hidden causal mechanisms.

8. Governance, Legitimacy, and Open Science

A technically executable observation is not automatically a legitimate observation. Provider terms, institutional governance, privacy, access conditions, and permissible retention can differ across services. MIBO-Education therefore uses fail-closed operational gates. Human records identify the protocol owner, document provider-specific terms review, and record an institutional ethics or governance determination. Agents may validate the records but may not approve themselves. A provider that is ineligible or unresolved produces planned missing cells; it does not cause silent substitution or retrospective removal from the registered panel.

The study currently involves API-based machine observations rather than human participants. That fact does not authorize an AI agent to declare ethics review unnecessary. The applicable institution must make and document the determination. Future Consumer Experience Mirror work or studies involving learners, teachers, or identifiable interaction logs will require separate protocols and review.

Open science is implemented through versioned artifacts rather than maximal disclosure without qualification. The public package contains the battery, codebook, scoring manual, observation protocol, scientific manifest, schemas, hashes, and code. Raw outputs are retained in an immutable private or controlled archive when provider terms, copyright, security, or institutional policy make unrestricted release inappropriate. Public metadata, hashes, coding, missingness, and analytical releases remain linked to L0 even when L0 itself cannot be distributed.

The MIBO-Education Protocol Package v1.0 is archived on Zenodo at https://doi.org/10.5281/zenodo.22047001. The package inherits the distinction established by MIBO Core between a frozen scientific specification and prospective execution records (Sasano, 2026a). GitHub preserves working history; Zenodo provides a citable, time-stamped release. Revisions receive new versions rather than overwriting the scientific history.

The observatory also adopts a corrigible claim lifecycle. Model comparisons are not published as permanent properties of brands. Every claim is scoped by service series, wave, item universe, environment, protocol version, and coding version. Later nonreplication, narrowed scope, amendment, or withdrawal remains part of the record. This is especially important when the object of study changes faster than conventional peer review.

Safeguard	Purpose
Frozen artifact hashes	Prevent silent changes to prompts, codebook, scoring manual, protocol, and manifest
Permanent series IDs	Preserve service continuity without confusing it with exact model identity
Exact wave model lock	Record what was requested and returned at the observation time
Fail-closed preflight	Block official execution when scientific, technical, governance, or credential gates are incomplete
Immutable L0	Preserve undesirable and unexpected responses as data rather than replacing them
Blind coding	Reduce model-reputation and wave-expectation effects on human judgments
Separate accuracy and behavior	Avoid treating correct task completion as sufficient educational quality
No overall winner score	Preserve construct trade-offs and bounded interpretation
Versioned corrections	Allow coding and claims to change without rewriting the historical observation

Table 4. Scientific and governance safeguards.

9. Discussion

MIBO-Education changes the unit of educational-AI evaluation. Most research asks whether an AI intervention improves a learner outcome, whether a model answers educational questions accurately, or whether a tutor follows pedagogical principles. Those questions remain necessary. The observatory asks an additional question that becomes unavoidable once general-purpose services are deployed at scale: what exactly was the educational actor at the time the evidence was produced, and did it remain the same educational actor later?

The framework's strongest conceptual contribution is the non-equivalence of capability progress and educational progress. Model developers may improve reasoning, context length, latency, or tool use while simultaneously changing how much work is returned to the learner, how uncertainty is expressed, or how assessment is conducted. Educational institutions cannot infer the latter from general capability leaderboards. They need domain-specific, repeated evidence.

The seven-dimensional fingerprint also places learner agency alongside conventional quality constructs. This choice is supported by an emerging literature in which generative AI can support autonomy, reflection, and self-regulated learning under some designs, but encourage cognitive outsourcing and overreliance under others (Yang et al., 2024; Zhai et al., 2024; Qian, 2025). The observatory does not presume that all cognitive offloading is harmful. It asks where the system draws the boundary between assistance and substitution in specific educational situations.

MIBO-Education is complementary to intervention research. A longitudinal machine panel can identify when a service's behavior changed; a learner study can test the consequences of that behavior. The two designs should eventually be linked. For example, if LAPI declines across waves, a separately reviewed human-participant study could examine whether learners show greater dependence or weaker unaided transfer. Conversely, intervention evidence can guide new Live supplements without altering the Frozen historical series.

The infrastructure also creates a basis for accountability in procurement and policy. An institution may adopt a named service after reviewing evidence about one configuration, only to receive a different configuration months later. A public educational-behavior trajectory does not replace local due diligence, but it can signal when prior evaluations should be repeated. OECD and UNESCO guidance both emphasize pedagogical purpose, human agency, safety, transparency, and continuing evidence in educational uses of generative AI (OECD, 2026; UNESCO, 2023). An observatory operationalizes part of that continuing evidence function.

Several extensions are intentionally outside W01. A Consumer Experience Mirror can observe public interfaces separately from first-party APIs and test fixed multi-turn scenarios without assuming interface equivalence. A bilingual Japanese–English mirror can examine functional language differences. A bounded Equity Sentinel can use counterfactual matched pairs in which only a construct-irrelevant learner cue changes. A Critical Event Overlay can screen low-frequency, high-severity events such as fabricated sources, dangerous instructions, authorship erasure, or false claims of authority. These extensions should receive separate manifests and datasets rather than being retrofitted into Frozen EBB-JA v1.0.

10. Limitations and Falsifiability

The first limitation is linguistic and cultural scope. EBB-JA v1.0 is a Japanese instrument. Results cannot be generalized automatically to English, other languages, or educational systems. Translation would create a new instrument requiring functional-equivalence work, not a clerical copy of the Japanese prompts.

Second, the panel is purposive. The selected series are prominent and methodologically diverse sentinels, not a probability sample of all AI services. A provider may be unavailable, terms-ineligible, or technically unqualified at a wave. Such absence limits comparison but is itself part of the deployment record.

Third, the primary condition is API-based and does not represent the full consumer experience. Public interfaces may include hidden instructions, personalization, tools, and routing absent from the research harness. MIBO-Education therefore does not claim API–UI equivalence. The future Consumer Experience Mirror is designed precisely because the two surfaces answer different questions.

Fourth, the item universe is bounded. Seventy situations cannot represent education in general, and domain means are not universal measures of teaching quality. The battery functions as a sentinel instrument: it is sufficiently broad to reveal multiple educational dimensions while sufficiently fixed to support longitudinal comparison.

Fifth, human coding remains interpretive. Item-specific anchors, blindness, training, double coding, and reliability reporting reduce but do not eliminate judgment. Model-assisted coding may support triage, but human calibration and adjudication remain necessary, especially for source verification, developmental appropriateness, and serious safety events.

Sixth, pedagogical drift is observational. It does not identify a provider's internal cause and does not prove consequences for learners. A stable result may be stably poor; an unstable result may include beneficial adaptation. The framework is falsifiable in a narrower sense: if coders cannot measure the constructs reliably, if within-cell variability overwhelms wave differences, if service-series continuity cannot be

defended, or if independent teams cannot reproduce the protocol, the corresponding claims must be narrowed or withdrawn.

11. Conclusion

General-purpose AI is not only entering education; it is changing while education uses it. That fact creates a measurement obligation. A one-time benchmark can document capability at a moment, and an intervention can estimate an effect under a defined treatment. Neither alone preserves the history of a continuing service whose educational behavior may change between evaluation and deployment.

MIBO-Education supplies that missing longitudinal layer. It follows persistent service series, fixes an educational instrument, samples conditional response distributions, separates accuracy from seven educational constructs, preserves immutable raw records, and treats operational and governance eligibility as part of the scientific design. Its founding proposition is simple:

AI capability benchmarks measure what a model can do. MIBO-Education measures what kind of educator a continuing AI service becomes.

The success of the observatory will not be established by dramatic rankings at Wave 1. It will be established if the same instrument can be administered, audited, re-coded, independently challenged, and used to revise claims when the services change. Wave 1 is therefore a baseline, not a verdict.

Declarations

Protocol and data status

This paper presents a frozen research design and reports no W01 comparative results. W01 is scheduled to begin on 1 September 2026 only after operational, provider, governance, model, schedule, site, and credential gates are satisfied. Engineering-pilot data are separated from official longitudinal data.

Ethics and governance

The current protocol concerns machine-only API observations and does not recruit human participants. An explicit institutional ethics or governance determination will be recorded before live qualification and official collection. Any future learner-, teacher-, or consumer-interface study will require a separate protocol and review.

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Competing interests

Kento Sasano is the chief executive officer and an equity holder of Scrumy Inc., a company operating in AI governance, AI security, and AI marketing. He also serves as an AI strategy advisor to Booost Inc. These relationships are disclosed because their activities may be perceived as relevant to AI evaluation and governance. The author retains responsibility for the research design, interpretation, and manuscript. No other competing interests are declared.

AI-assistance disclosure

Generative-AI systems assisted with literature discovery, research organization, language editing, document production, structural critique, and consistency checking. The author selected the research design, verified cited sources and project records, made all substantive decisions, and accepts responsibility for the manuscript. Generative-AI systems are not authors.

Materials availability

The working implementation is maintained at https://github.com/mibo-research/mibo-satellite/tree/main/satellites/mibo-education. The frozen MIBO-Education Protocol Package v1.0 is archived on Zenodo at https://doi.org/10.5281/zenodo.22047001. MIBO Core Protocol Package v1.0 is archived at https://doi.org/10.5281/zenodo.21936410.

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Appendix A. Frozen Scientific Artifact Hashes

The following values identify the authoritative scientific source integrated into the MIBO-Education implementation. A byte-level mismatch must fail scientific validation.

Artifact	SHA-256
EBB-JA v1.0	5475d4c0edd651ae4fbc6e79c42642433fa7e0c27dba616a972236d35517cd4e
Codebook v1.0	c38cce32bad4e3b156ec6094d2368fca0937536519b25a2a80e115042da9161a
Item-Level Scoring Manual v1.0	3ae057c2e5f23312b5ca000f3d9f42ced4db23ca6b6db8c4889905c9f9a0bbef
Observation Protocol v1.0	ca9b51ecbca1f5fd61443e7921ff4305dac82829a3043c795416c437b20d5d8f
W01 Scientific Manifest v1.0	ecb5b5aa779d0806cb034581d76ab1c128587f1b9590a7b482fc3fa93336a6f1
External prompt lock	8e251463eebb1c2e3f86ae7e82d98a4685afbf2dd3434731cdc72939d6ef6a82

Appendix B. W01 Design Summary

Design element	W01 scientific specification
Instrument	EBB-JA v1.0; 70 Frozen Japanese items; E1-01 through E7-10
Primary environment	CLOSED for M01–M04: no tools, web, retrieval, RAG, files, memory, or history
Native mirror	M05 NATIVE; analysed separately and not pooled as an equivalent environment
Session	Independent single turn; no researcher-added system or developer prompt
Replication	k = 10 per service × item × wave cell
Sampling	Provider-native defaults unless a value is required; no optional reasoning override
Output cap target	8,192 visible-output tokens, with provider-specific documentation
Execution	Stratified block-randomized interleaving; replications distributed through the wave
Wave window	12-hour target; 24-hour hard window; canonical timestamps in UTC
Data chain	L0 immutable raw → L1 canonical → L2 coded → L3 analytical
Retry	Transport failures only; refusals, blocks, hallucinations, truncation, and poor answers are data